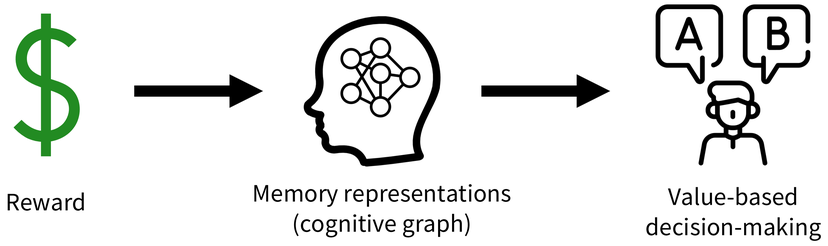
Reward restructures existing cognitive maps

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Department of Psychology[1], Institute for Mind and Biology[2], Neuroscience Institute[3], The University of Chicago

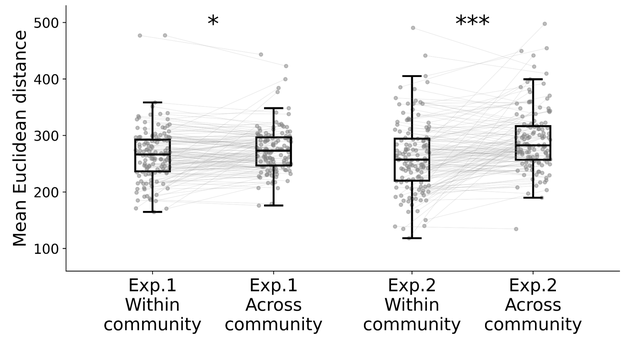
seoyounglee@uchicago.edu

How does reward affect memory representations and decision-making?



Does reward spread in a graded manner throughout a community-structured cognitive graph and influence value-based decision-making?

Participants learned the community structure

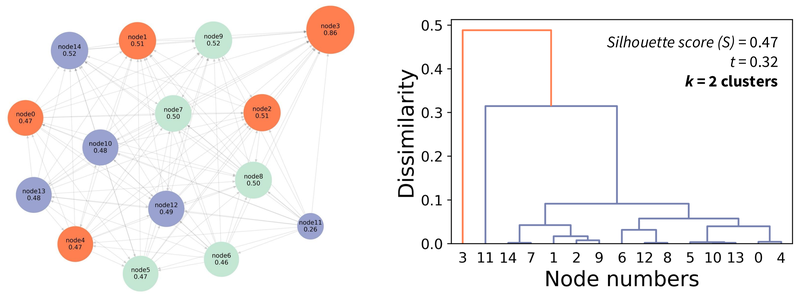


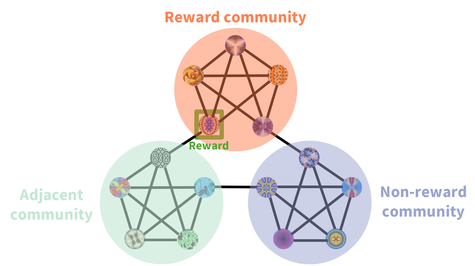
Participant graph reconstruction examples:



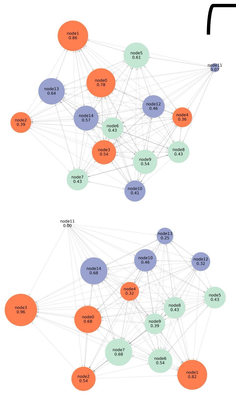
Learned community-structured graph was reorganized

Average reconstructed graph across participants based on choice



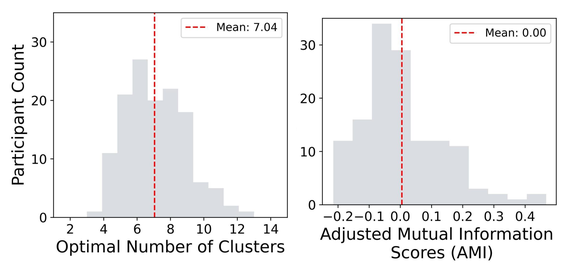


Learned community-structured graph



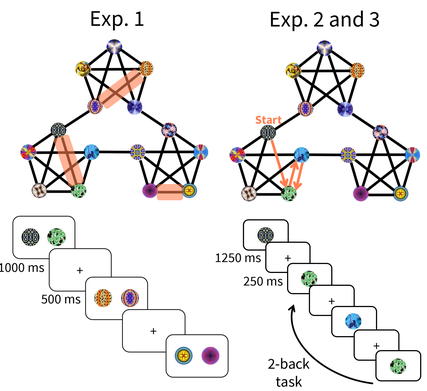
Examples of participants' reconstructed graphs based on choice

Participant-level reconstructed graphs based on choice

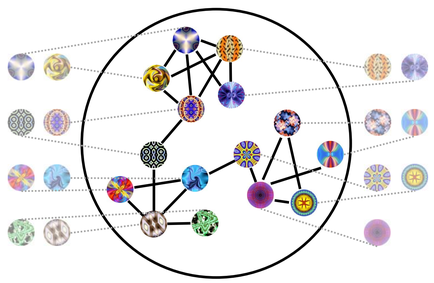


Methods

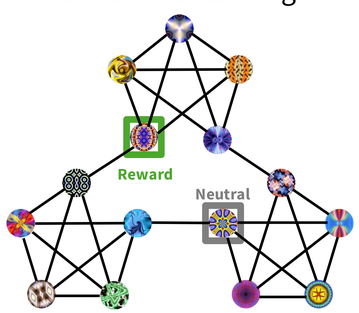
1. Community-structured graph learning



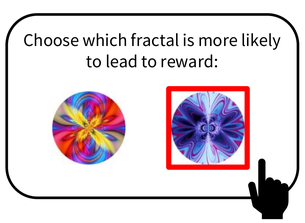
2. Graph test



3. Reward learning



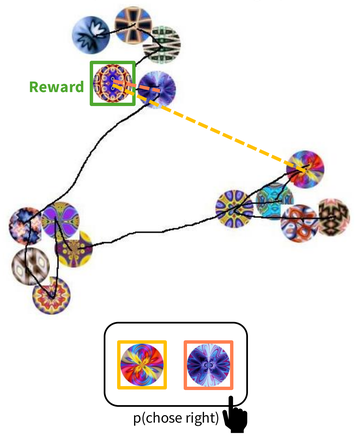
4. Decision-making



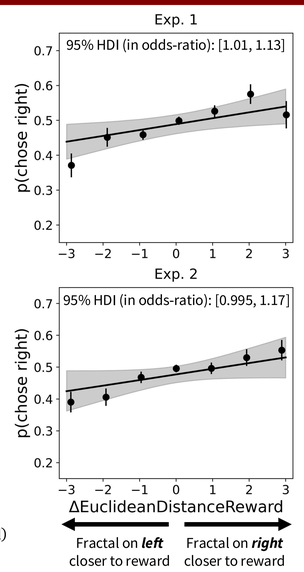
Distance from reward influences decisions

(only when structure is explicitly remembered)

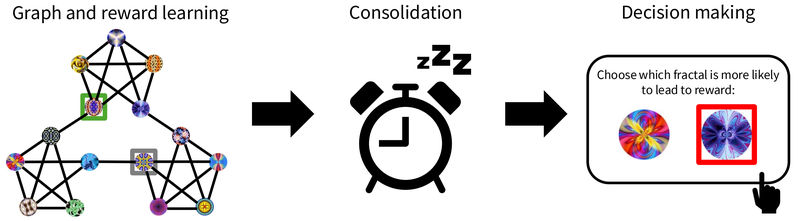
Participant reconstructed graph:



$$\Delta EuclideanDistanceReward = d(\text{rewarded fractal, left fractal}) - d(\text{rewarded fractal, right fractal})$$



Future directions



Conclusion

Explicit memory representations enable reward to spread broadly across association space. Without them, reward actively reshapes associative memory structures to support value-based decision making.

Preprint: